PROJECT TOPIC:

MR KLEAN LAUNDRY BUSINESS:

FROM DATA TO DECISION

1. Introduction

Imagine you own a busy laundry business. Every day, customers bring in clothes for washing, dry cleaning, ironing, stain removal, and mending. As the business grows, hundreds of orders are processed, payments are received through different methods, employees handle numerous tasks, and machines are used continuously.

After several months, thousands of records have accumulated. Although the business has plenty of data, management s till struggles to answer important questions:

- Which customer categories (Walk-in, Regular, Corporate, and Student) generate the highest revenue and net profit?
- Do customers from different states contribute differently to revenue and profitability?
- Which laundry services generate the highest revenue and profit?
- Is there a relationship between the quantity of clothes, the weight of clothes processed, and the revenue generated?

1. Introduction (Continued)

- Which delivery option (Customer Pickup, Home Delivery, or Express Delivery) contributes most to business revenue?
- Which operational cost components contribute most to the overall expenses incurred by the business?
- Are there monthly or seasonal patterns in customer demand, revenue generation, and profitability?
- Which factors appear to have the greatest influence on the profitability of the business?

2. Project Objectives

To transform business data into actionable insights that support strategic and operational decision-making, while building an ensemble machine-learning model to predict Net Profit.

3. Data Description

The Mr Klean Laundry Business Data was renamed as LaundryData and the structure is as follows:

- 1,000 observations and 32 variables.
- 18 numerical variables
- 8 categorical variables
- 4 date variables
- 2 identifier variables
- Predictive target: NetProfit.

4. Source and Design of the Data

The dataset is a realistic synthetic business dataset created for the Purpose this Mr Klean Laundry Business project.

The relational business structure was designed in MySQL where tables and attributes were created.

List of tables that were created includes customers, orders, services, employees, machines, assignments, payments and expenses.

Artificial or synthetic records were generated Mockaroo using the tables created in MYSQL.

Data was imported into MySQL for validation, cleaning, transformation and integration.

The integrated dataset was exported as a CSV file for analysis in R.

5. Data Pre-processing

Duplicate check: examined the dataset for repeated records using this code anyNA(LaundryData).

Missing-value check: checked missing observations using this code is.na(LaundryData) and determine appropriate treatment.

Outlier detection: numerical variables like revenue, total expenses, net profit, customers age, quantity of clothes, weight of clothes were all inspected for outlier but only ‘weight of clothes’ show outlier with unusually high or low observations.

Data transformation: convert variables to appropriate data types and prepare categorical variables for modelling.

Scaling/standardization: apply where required by the modelling technique.

Target preparation: define NetProfit as the response variable for predictive modelling.

The supplied presentation documents the preprocessing workflow but does not provide numeric duplicate, missing-value or outlier counts; no counts are invented here.

5. Data Pre-processing — Outlier Detection

DATA PREPARATION

The boxplot shows the distribution of WeightKg and identifies observations beyond the upper whisker as statistical outliers.

Most observations fall approximately within the 4–11 kg range; the upper whisker extends to about 22 kg.

Several observations above 22 kg extend to approximately 32 kg and are flagged as outliers.

These observations were retained because unusually heavy laundry orders can represent valid high-volume commercial transactions rather than data-entry errors. Outliers was not detected in other variables.

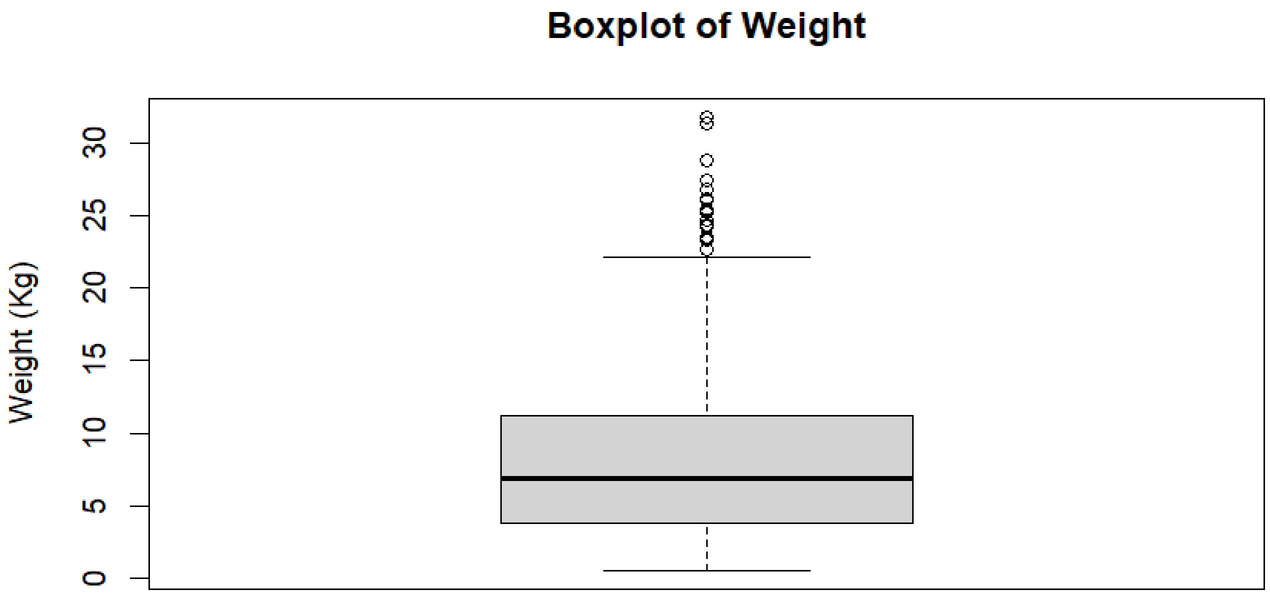

Fig. 1. Boxplot of Weight (Kg): Outlier Detection

6. Exploratory Data Analysis

EDA was used in this project to understand the structure and behaviour of the business data based on the management questions before modelling.

Univariate analysis examined individual variables and distributions.

Bivariate analysis examined relationships between two variables.

Multivariate analysis compared multiple measures across business categories.

Bar charts, trend charts and correlation analysis were used to communicate management-relevant findings.

7. Customer Category Analysis

Management question 1.: Which customer categories generate the highest Revenue and NetProfit?

Customer Category is compared with two financial measures, making this a multivariate bar chart.

Regular customers are the strongest contributor in the documented chart.

Business use: support customer retention, acquisition and segment prioritization.

Fig. 2. Revenue and Net Profit by Customer Category

8. State Contribution Analysis

Management question 2.: Do customers from different states contribute differently to revenue and profitability?

Revenue and NetProfit were compared across customer states.

The horizontal format improves readability when many state names are displayed.

Business use: identify stronger markets and areas requiring targeted marketing or service improvement.

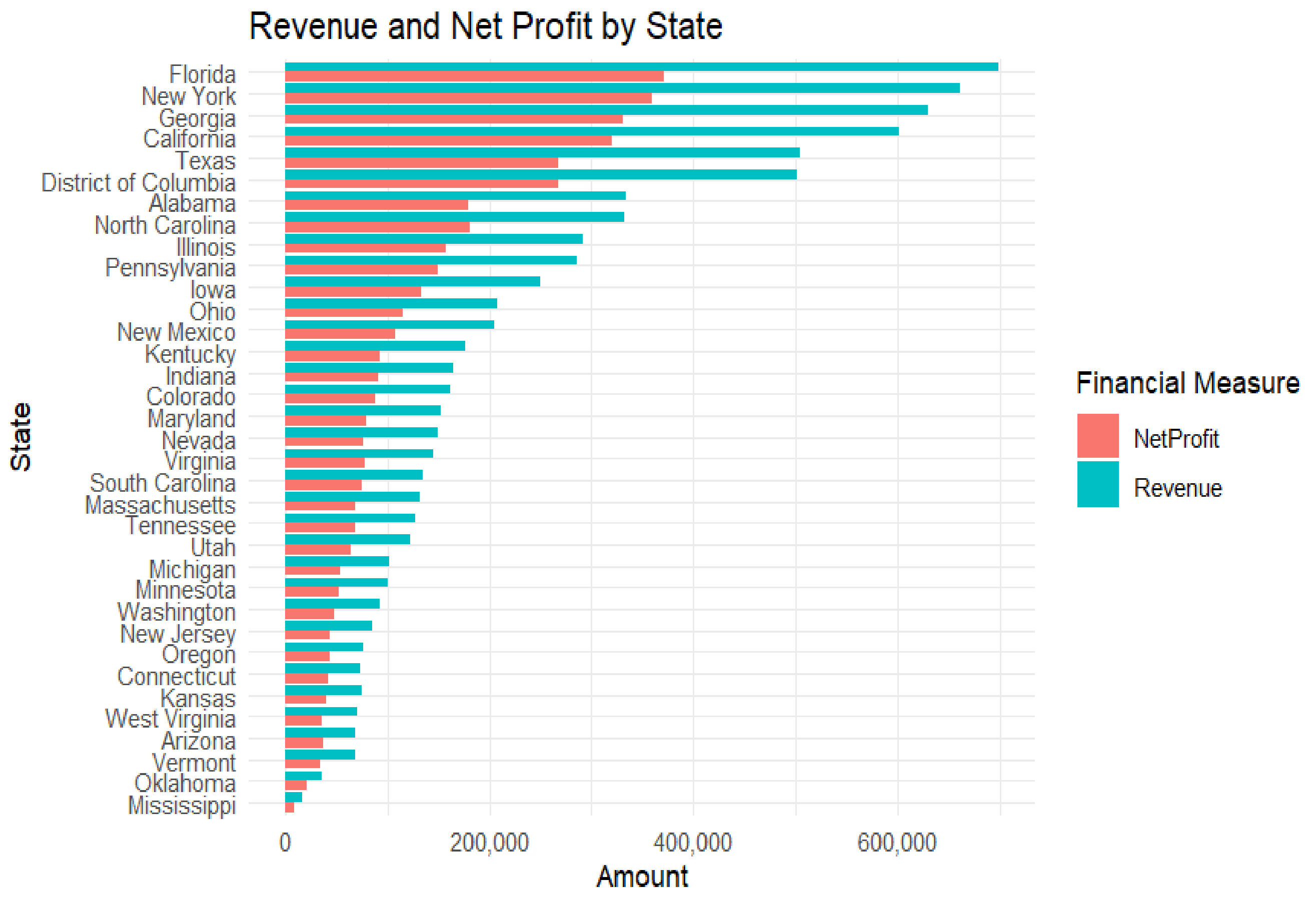

Fig. 3. Revenue and Net Profit by State

9. Service Revenue and Profitability

Management question 3.: Which laundry services generate the highest revenue and profit?

Laundry recorded the highest total revenue and net profit.

Revenue: $3,862,076.

Net Profit: $1,959,697.

Business use: protect capacity and service quality in strong services while reviewing weaker services.

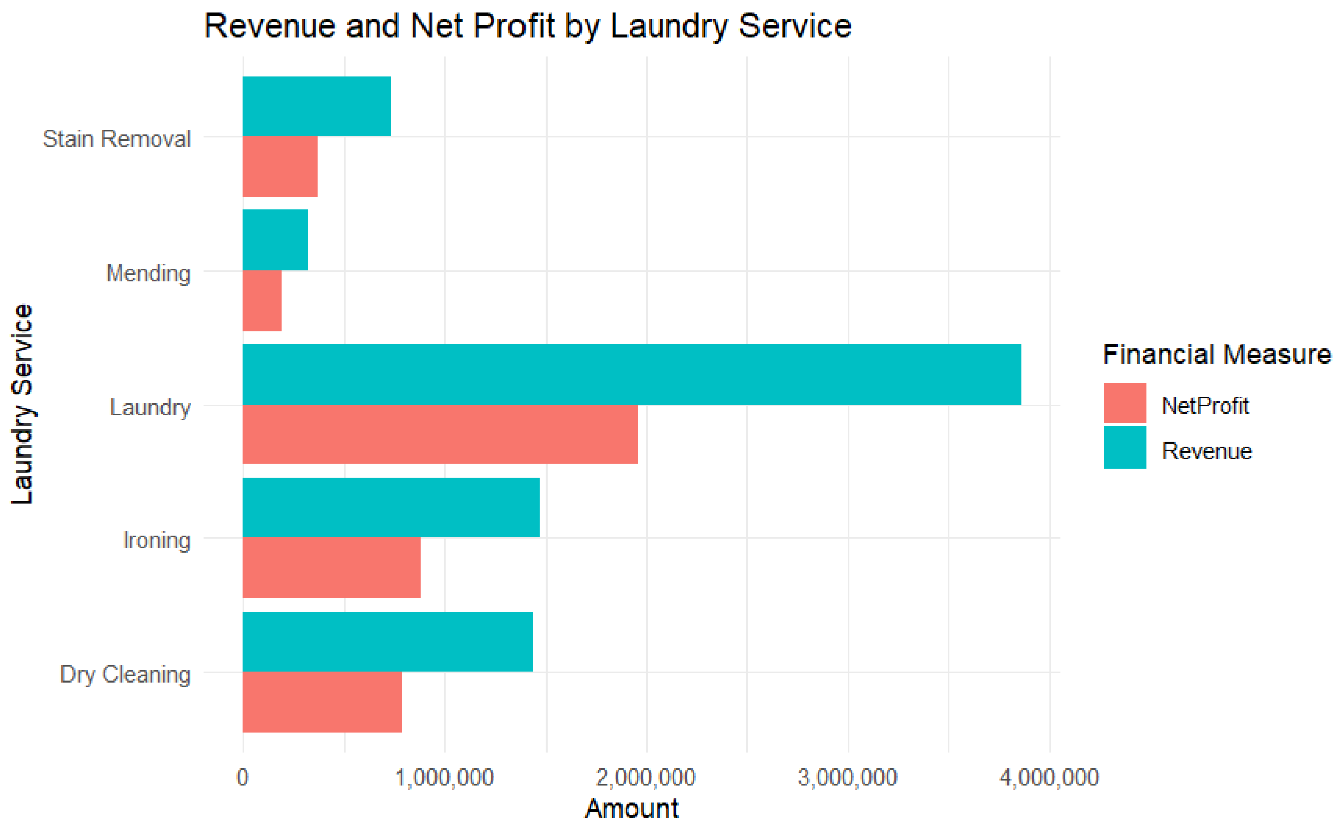

Fig. 4. Revenue and Net Profit by Laundry Service

10. Quantity, Weight and Revenue

Management question 4.: Is there a relationship between the quantity of clothes, the weight of clothes processed, and the revenue generated?

Quantity ↔ Weight: r = 0.753 — strong positive relationship.

Quantity ↔ Revenue: r = 0.009 — virtually no linear relationship.

Weight ↔ Revenue: r = 0.024 — virtually no linear relationship.

Conclusion: quantity and weight alone do not explain revenue but both are strongly related.

Business use: consider service type, pricing, customer and delivery factors.

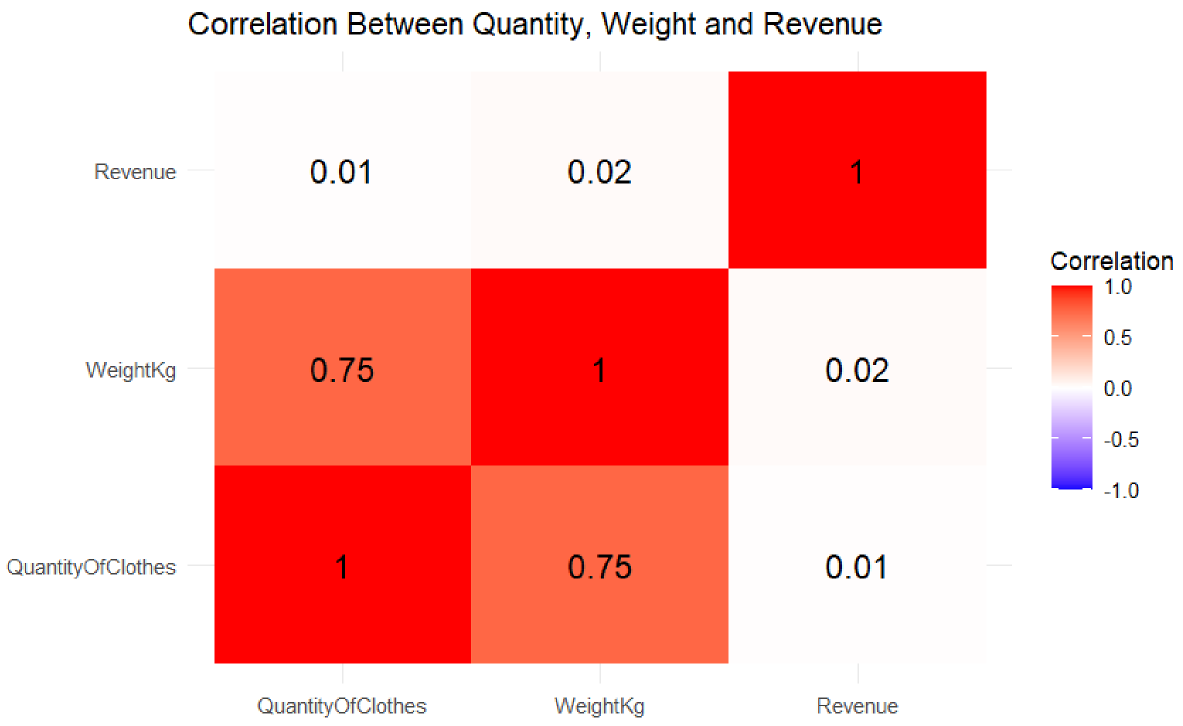

Fig. 5. Relationship Between Quantity, Weight and Revenue

11. Delivery Option Analysis

Management question 5.: Which delivery option contributes most to revenue and profitability?

Customer Pickup is the strongest delivery option.

Revenue: $4.55M.

Net Profit: $2.51M.

Home Delivery is second; Express contributes the least.

Business use: maintain strong pickup capacity and review the economics of other options.

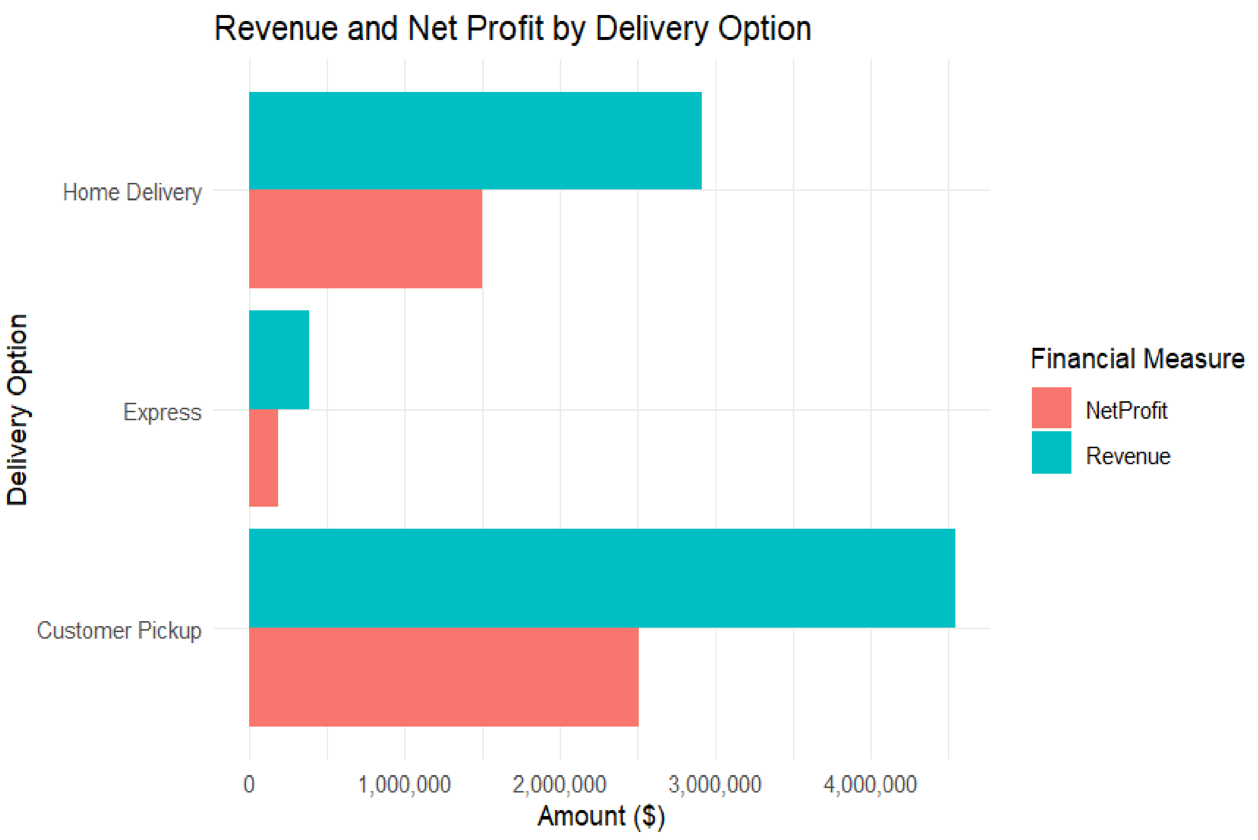

Fig. 6. Revenue and Net Profit by Delivery Option

12. Operational Cost Analysis

Management question 6.: Which operating costs contribute most to overall expenses?

Labour Cost is the dominant expense: $1.63M.

Tax: $587,944; Electricity: $390,188.

Maintenance and Detergent are also significant.

Business use: focus cost-control efforts on the largest cost drivers.

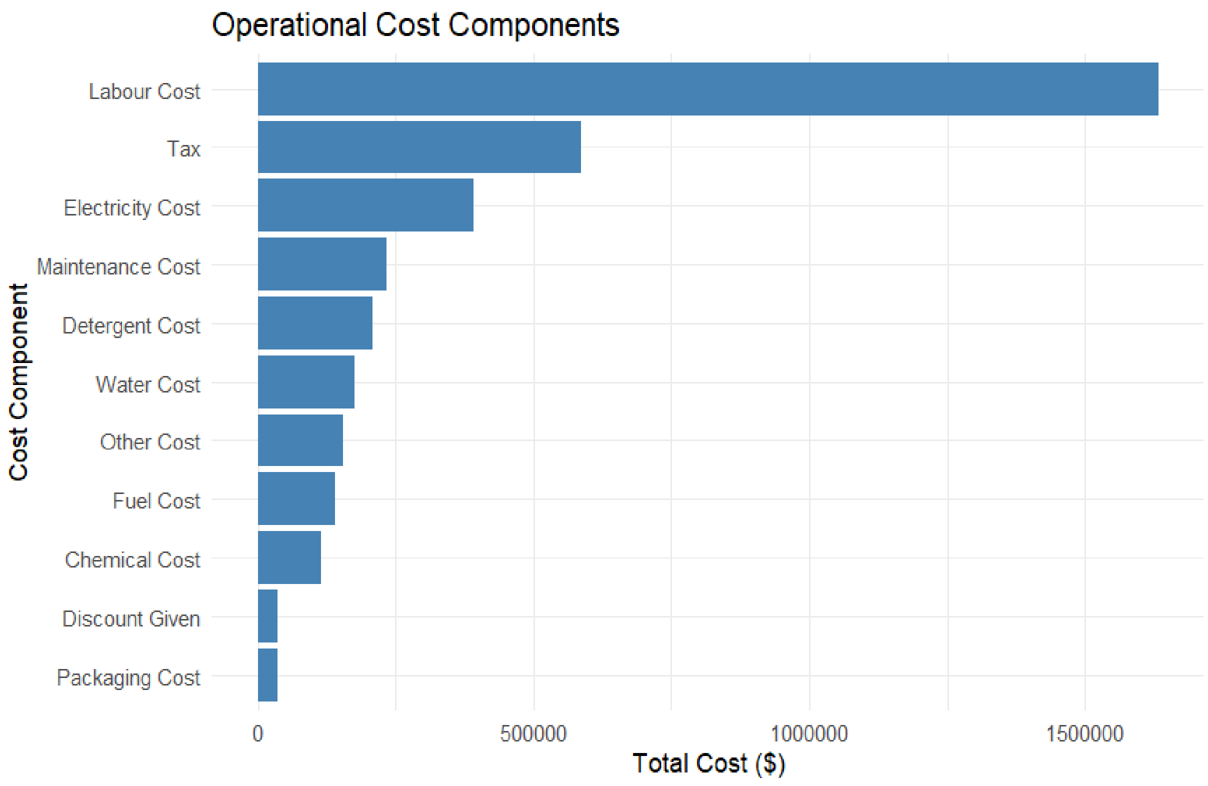

Fig. 7. Operational Cost Components

13. Monthly Demand / Order From 2021 to 2025

Management question 7.: Are there monthly or seasonal patterns in customer demand, revenue generation, and profitability?

Monthly demand fluctuates substantially.

High-demand months generally produce higher revenue and profit.

No consistent recurring seasonal pattern was established across the documented period.

Business use: use flexible capacity planning and monitor high- and low-demand periods.

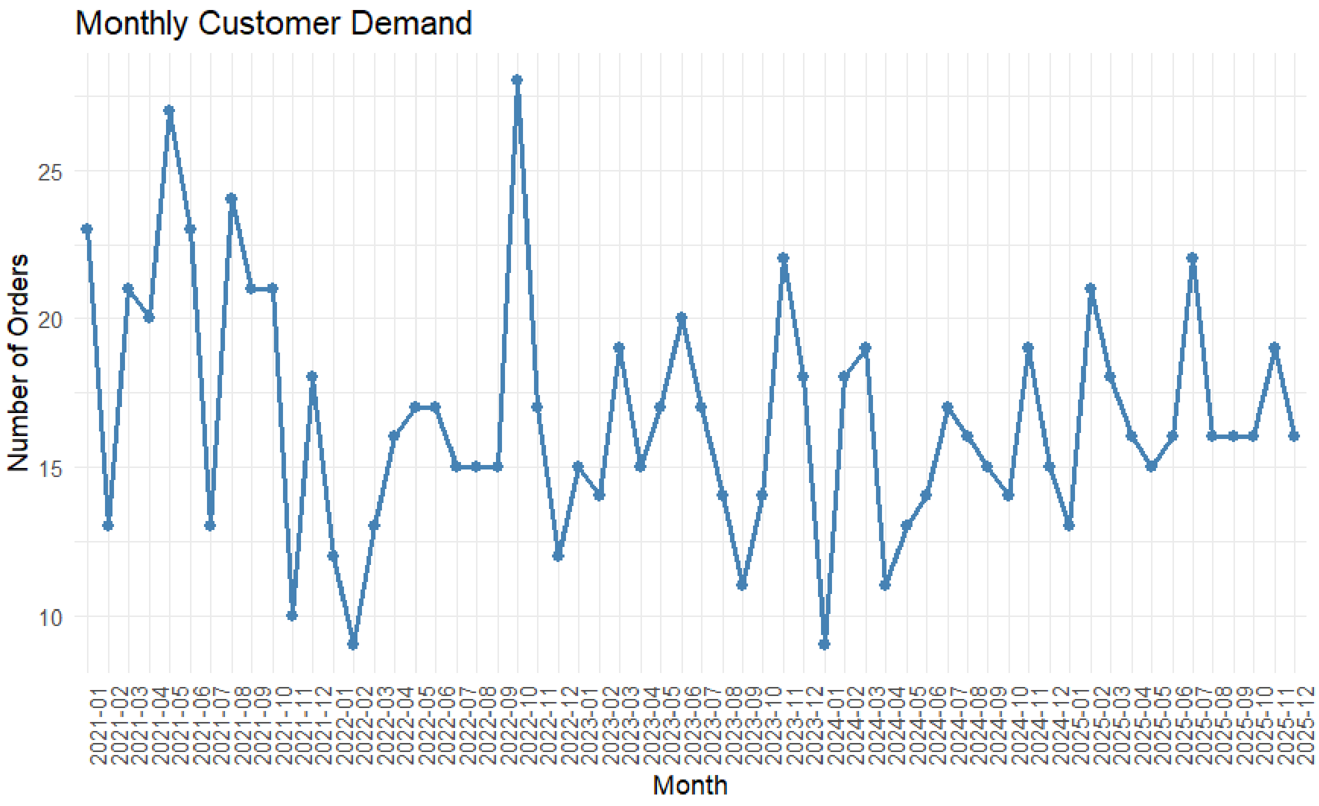

Fig. 8. Monthly Customer Demand

14. Monthly Trend Revenue From 2021 to 2025

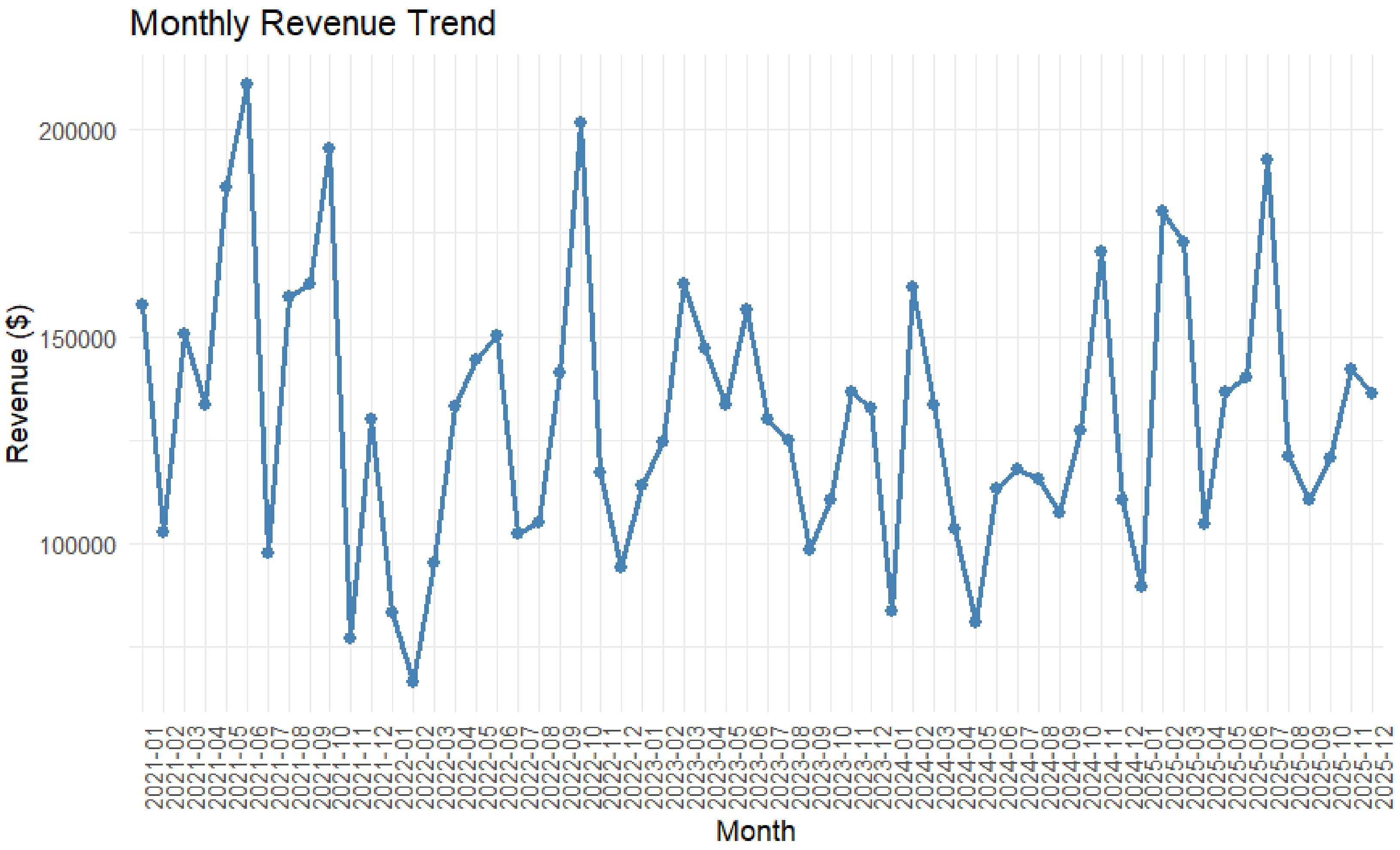

Fig. 9. Monthly Revenue Trend

Revenue shows irregular peaks and troughs across the documented months.

The pattern supports active monthly monitoring rather than assuming a fixed seasonal cycle.

Revenue and NetProfit generally move in the same direction.

15. Correlation

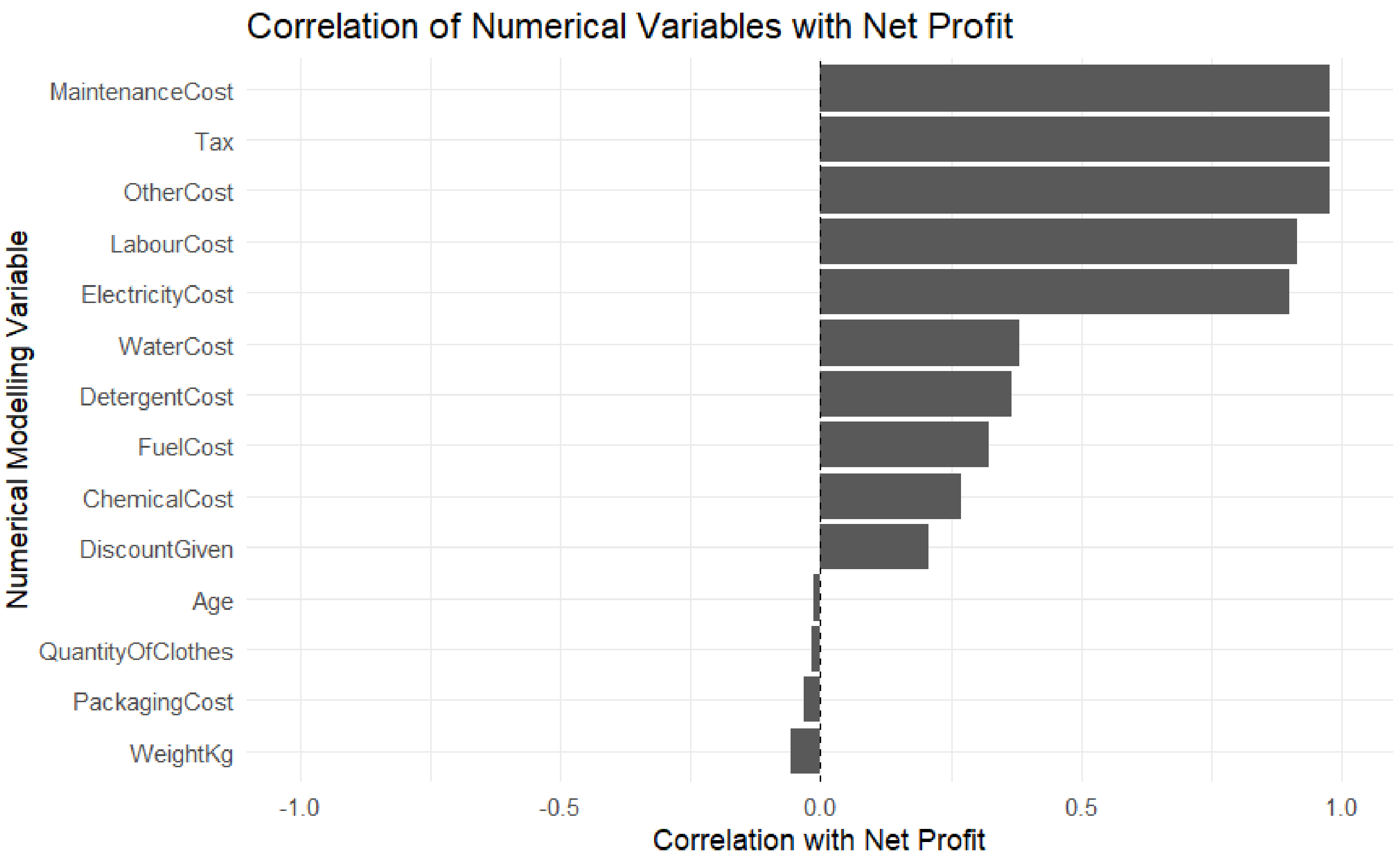

Fig. 10. Correlation Matrix of Documented Numeric Relationships

Correlation analysis was carried out to understand the relationship between the numerical modelling variables and Net Profit before building our predictive models.

The analysis showed that some cost variables, particularly Maintenance Cost, Tax, Other Cost and Labour Cost, had strong positive relationships with Net Profit, while variables such as Age, Quantity of Clothes and Weight had weak linear relationships.

This analysis provided an initial understanding of the data and helped guide our subsequent feature-importance and modelling process.

16. Feature Selection / Importance

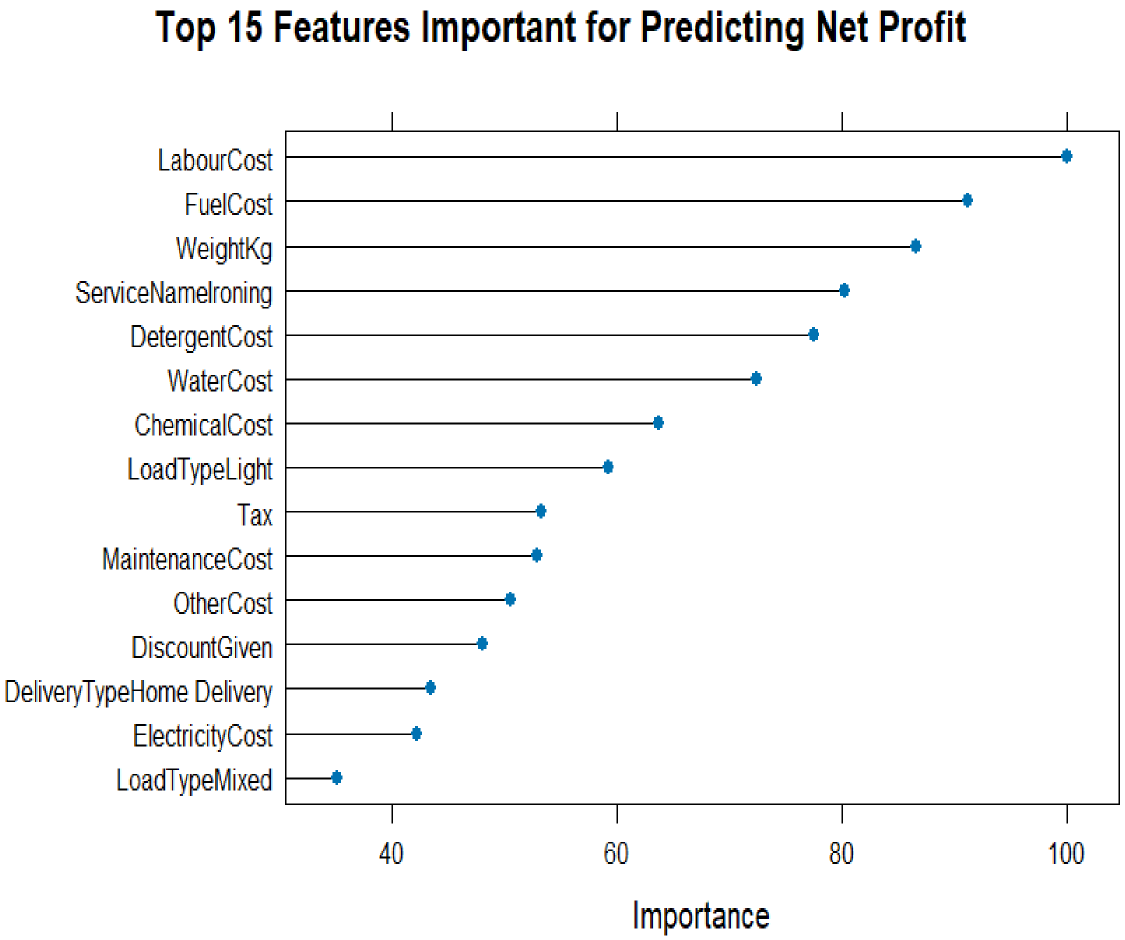

Fig. 11. Feature Importance for predicting Net Profit

After carrying out the correlation analysis, we performed feature importance analysis to identify which variables contribute most to predicting Net Profit in our machine-learning model.

The chart ranks the top 15 features according to their importance to the model and the most important features shown are:

1. LabourCost — highest importance, approximately 100
2. FuelCost — approximately 91
3. WeightKg — approximately 87
4. ServiceName: Ironing — approximately 80
5. DetergentCost — approximately 78
6. WaterCost — approximately 72
7. ChemicalCost — approximately 64
8. LoadType: Light — approximately 59
9. Tax — approximately 53
10. MaintenanceCost — approximately 53

The remaining features have lower importance but still contribute to the model

17. Ensemble Modelling

- Six different predictive models were built and evaluated to predict Net Profit.
- The data was divided into 80% training data and 20% testing data.
- Set seed (123) was used to ensure that the results could be reproduced.
- 10-fold cross-validation was used to train and compare all models under the same conditions.
- The best-performing model was selected based on RMSE (Root Mean Squared Error), R^2 (R-Squared), and MAE (Mean Absolute Error).
- Models Built are
- PLS — Partial Least Squares
- GBM — Gradient Boosting Machine
- RF — Random Forest
- SVM — Support Vector Machine
- kNN — k-Nearest Neighbours
- Bagging — Bootstrap Aggregating

18. Actual vs Predicted Net Profit Across Six Machine Learning Models

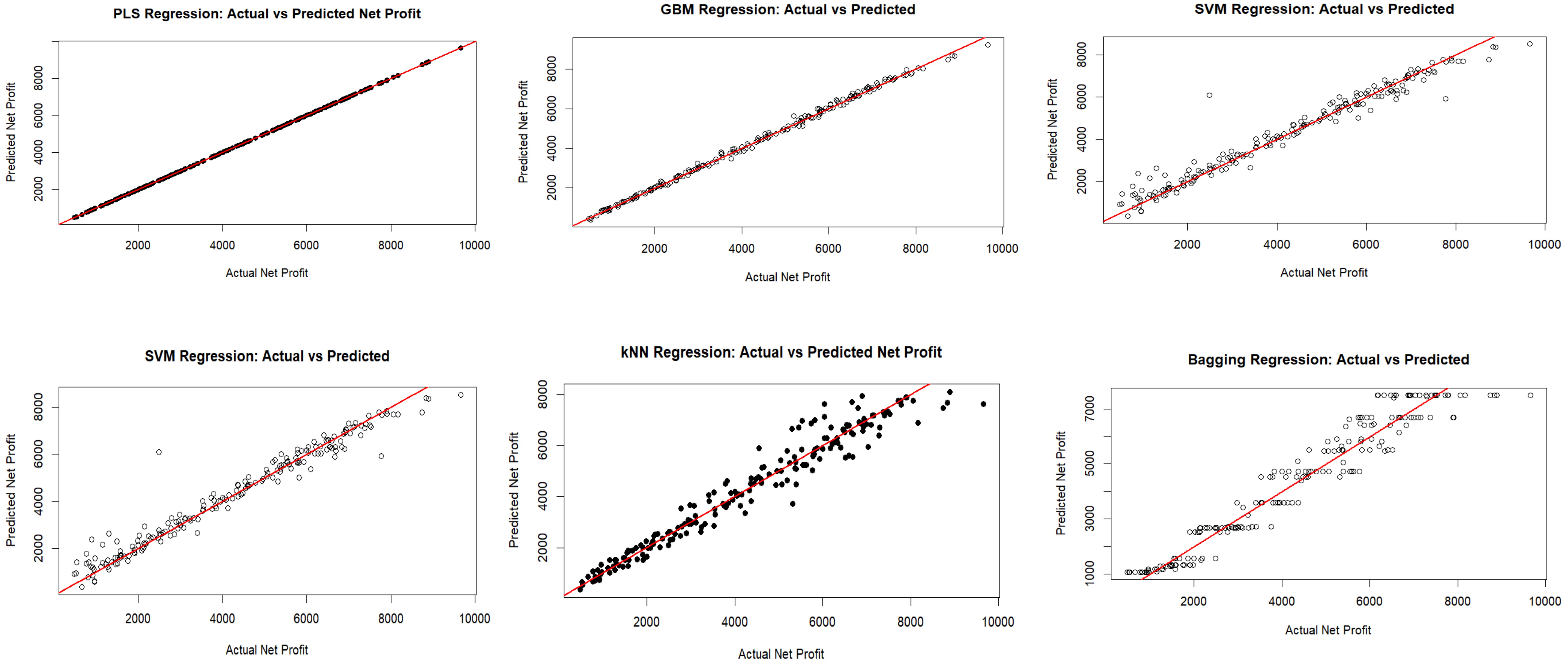

Fig. 12. Actual vs Predicted Net Profit: Model Comparison

19. Six-Model Performance Comparison

MODELLING

Model	RMSE ↓	R^2 ↑	MAE ↓	Performance
PLS (Partial Least Squares)	0.0336	1.0000	0.0280	Excellent
GBM (Gradient Boosting Machine)	125.0357	0.9907474	90.0455	Very Good
Random Forest	300.1462	0.9827295	215.2852	Very Good
SVM (Support Vector Machine)	450.6877	0.9622112	270.0380	Good
kNN (k-Nearest Neighbours)	405.5169	0.9515488	334.1182	Good
Bagging (Bootstrap Aggregating)	535.7497	0.9464361	421.0446	Fair

Interpretation: lower RMSE and MAE indicate smaller prediction errors, while higher R^2 indicates that more variation in NetProfit is explained.

PLS ranks first in the recorded comparison, followed by GBM and Random Forest. Bagging records the weakest overall performance.

20. Model Performance Evaluation and Final Model Selection

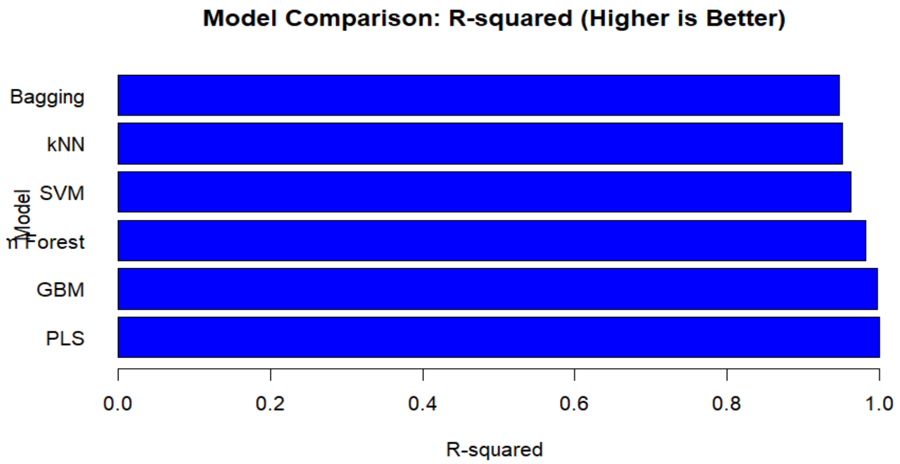

Fig. 11. Model Comparison by R-squared

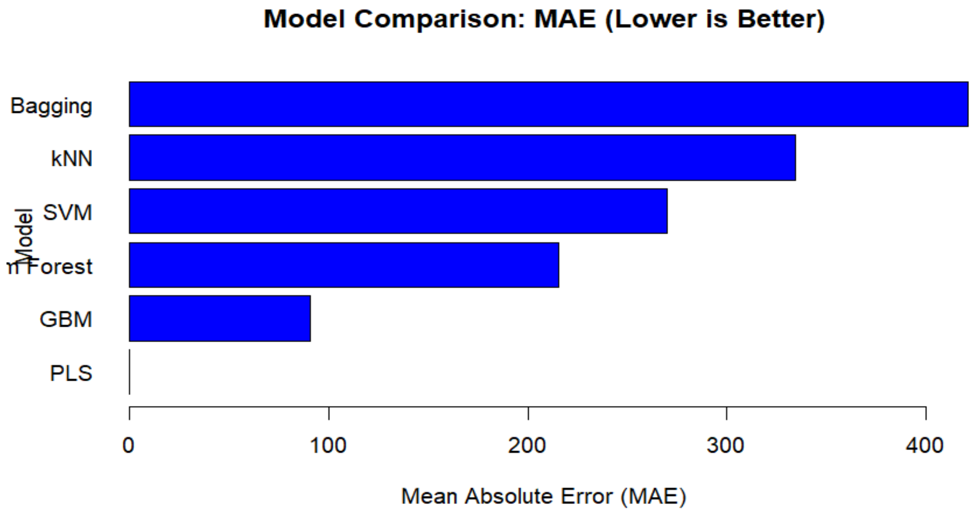

Fig. 12. Model Comparison by MAE

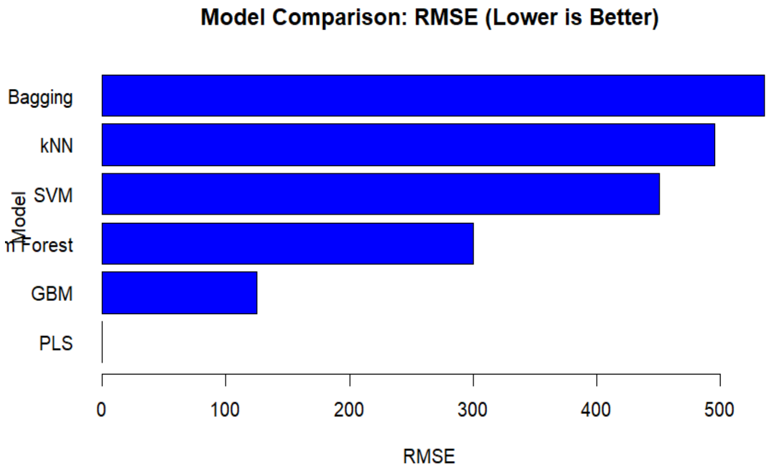

Fig. 13. Model Comparison RMSE

PLS was not selected as the final model based on one metric alone. All six models were compared using R-squared, RMSE (Root Mean Squared Error)and MAE (Mean Absolute Error).

PLS consistently produced the best results: the highest R-squared and the lowest prediction errors. Therefore, PLS was selected as our final model for Net Profit prediction."

21. Overall Findings

- Regular customers are a major contributor to Revenue and NetProfit in the documented customer analysis.
- Laundry is the strongest-performing service in the documented service analysis.
- Customer Pickup is the strongest delivery option.
- Labour Cost is the largest operational expense.
- Quantity and Weight are strongly related, but neither is meaningfully correlated with Revenue.
- Monthly performance fluctuates substantially without a clearly established recurring seasonal pattern.
- LabourCost has the highest recorded PLS variable importance.
- PLS ranks first among the six recorded predictive models.

22. Recommendations

- Improve staff scheduling, workload distribution and labour productivity.
- Monitor Labour, Electricity, Maintenance and Detergent costs closely.
- Maintain strong Customer Pickup capacity and service quality.
- Use customer and service profitability analysis to guide retention, marketing and capacity decisions.
- Do not rely on quantity or weight alone for pricing or revenue expectations.
- Use monthly monitoring to respond early to weak and high-demand periods.
- Continuously validate and retrain the predictive model as new business data becomes available.

23. Conclusion

The project transformed integrated Mr Klean Laundry data into actionable business intelligence using R.

The analysis combined preprocessing, EDA, correlation analysis, feature importance and predictive modelling.

The major management priorities are labour efficiency, cost control, customer profitability, service performance and delivery operations.

The documented modelling comparison selected PLS as the best-performing model.

The project establishes a repeatable framework: structured data → analysis → insight → prediction → action.

THANK YOU